

Flexible Nuisance Learning for D-IV-LATE: Runtime Gains, Inference Limits, and Model Choice

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Abstract

We study how the choice of nuisance learner affects estimation of the Distributional Instrumental Variable Local Average Treatment Effect (D-IV-LATE) within a double/debiased machine learning framework. We compare a Random Forest benchmark against Kolmogorov-Arnold Networks (KANs) and organize the paper around three evidence blocks: a corrected estimand-aligned simulation benchmark, a targeted KAN ablation, and an empirical application to the effect of 401(k) participation on the distribution of net financial assets.

The simulation benchmark uses explicit monotone principal strata and a fixed complier truth bundle across 27 scenarios spanning three design families, three instrument-strength regimes, and three sample sizes. Under the corrected estimator and locked width-64 KAN policy, KANs are faster in all 27 scenarios but have lower integrated RMSE in only one scenario; Random Forests have lower RMSE in the remaining 26. A targeted KAN ablation selects the width-64 KAN configuration as the best KAN variant among the configurations tested, but the model-comparison evidence is a speed/accuracy tradeoff rather than a KAN accuracy-dominance result. The empirical application shows that nuisance-model choice remains substantively important: RF and KAN estimates differ materially in the core specification, with frequent sign reversals along the estimated D-IV-LATE curve. However, the empirical KAN propensity estimates concentrate near the boundaries, so the empirical KAN curve is treated as model-sensitivity evidence rather than as a validated standalone causal curve.

The central limitation is now inferential rather than identification-theoretic. This revision replaces the earlier residualized score with a reduced-form orthogonal level-score estimator for the conditional-IV D-IV-LATE target and regenerates the main numerical evidence under that corrected code path. The resulting conclusion is mixed. KANs provide a clear runtime advantage in the implemented benchmark, but they do not deliver broad point-accuracy gains and they are not validated as a default inferential layer.

Keywords D-IV-LATE, Instrumental Variables, Double Machine Learning, Kolmogorov-Arnold Networks, Random Forests, Model Choice

1. Introduction

Understanding treatment effects across the entire outcome distribution is often more informative than focusing on a single average effect. In instrumental-variables settings, this leads naturally to the Distributional Instrumental Variable Local Average Treatment Effect (D-IV-LATE), which traces how an endogenous binary treatment shifts the outcome distribution for compliers [13, 1]. The double/debiased machine learning (DML) framework is attractive in this setting because it permits flexible nuisance estimation while retaining orthogonality-based protection against first-order regularization bias [8].

What DML does not settle is the practical choice of nuisance learner. That choice is often treated as secondary, but in distributional IV problems it can affect both the point estimate and the stability

of the implied inferential procedure. This paper studies that question directly by comparing a Random Forest benchmark to Kolmogorov-Arnold Networks (KANs), a spline-based neural architecture motivated by the Kolmogorov-Arnold representation theorem [17, 15]. The relevant question is not whether KANs are universally superior. It is when they improve D-IV-LATE point estimation or runtime under a disciplined benchmark, and whether any such gains survive contact with uncertainty quantification.

Our evidence is organized around three components. First, we rebuild the simulation benchmark so that the reported truth is a genuine complier D-IV-LATE generated from explicit monotone principal strata. The resulting full matrix spans three design families, three instrument-strength regimes, and three sample sizes. In that benchmark KAN is faster than Random Forests in all 27 scenarios, but RF has lower integrated RMSE in 26 of 27 scenarios. The lone KAN RMSE win occurs in the `complex.local--strong--n2000` cell, where RF exhibits a large integrated-RMSE tail instability even though its mean absolute error remains slightly lower. Second, we run a targeted KAN ablation and select a width-64 KAN as the best KAN point-estimation variant among the tested configurations. Third, we revisit the 401(k) application and find that the nuisance-model choice remains substantively important in the data: RF and KAN estimates differ materially in the corrected core specification.

The paper’s main result is therefore not a blanket endorsement of KANs. A dedicated inference validation study shows that KAN-based pointwise intervals undercover badly under both asymptotic and bootstrap constructions. Random Forest intervals are materially better calibrated in the validation design. The strongest defensible conclusion is thus mixed: KANs are computationally attractive nuisance learners in this implementation, but they are not yet validated as an end-to-end point-estimation or inferential default.

This paper makes four contributions. First, it provides a corrected and reproducible benchmark for D-IV-LATE model comparison. Second, it documents a clear computational speed advantage for the implemented KAN nuisance learner, while showing that this advantage does not translate into broad RMSE gains in the corrected benchmark. Third, it shows that nuisance-model choice remains substantively important in an empirical application. Fourth, it demonstrates that point-estimation performance, runtime, and inferential validity need not move together, and that this separation matters for what can be claimed credibly in applied work.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 presents the identification framework for D-IV-LATE, and Section 4 describes the orthogonal-score estimator and the nuisance-learning implementations. Section 5 states the asymptotic result. Section 6 evaluates uncertainty quantification, Section 7 reports the simulation benchmark and KAN ablation, Section 8 presents the 401(k) application, Section 9 discusses the implications for model choice, and Section 10 concludes.

2. Literature Review

Our research is situated at the confluence of several major streams of literature. These include the estimation of distributional treatment effects (DTEs), the use of instrumental variables (IV) to address endogeneity, modern machine learning (ML) methods for causal inference, and the emerging field of Kolmogorov-Arnold Networks (KANs). This paper synthesises these areas, particularly by exploring the integration of KANs into the double/debiased machine learning (DML) framework for estimating the Distributional Instrumental Variable Local Average Treatment Effect (D-IV-LATE).

First, our work is grounded in the extensive literature on estimating treatment effects beyond simple averages. The foundational idea that policy evaluation should consider the entire distribution of outcomes dates back to seminal work on Quantile Treatment Effects (QTEs) by Doksum [9] and Lehmann & D’Abrera [16]. This field was significantly advanced by Koenker [14] with the development of quantile regression,

and Firpo [10] further contributed methods for estimating quantile effects under endogeneity. More recent research has shifted focus towards directly estimating the DTE, which captures the impact on the cumulative distribution function (CDF), as seen in Chernozhukov et al. [7]. Our paper follows this distributional path, extending it to settings with endogenous treatments.

Second, we connect with the classic literature on instrumental variables. The use of an instrument to address endogeneity was formalised by Angrist et al. [2], who introduced the Local Average Treatment Effect (LATE). This parameter identifies the average causal effect for the subpopulation of 'compliers' whose treatment status is influenced by the instrument. The distributional analogue of the LATE was subsequently explored by Imbens & Rubin [13] and Abadie [1], who laid the groundwork for understanding distributional effects in an IV setting. This paper focuses on developing and comparing robust estimators for this specific distributional IV parameter.

Third, our estimation strategy is built upon the DML framework pioneered by Chernozhukov et al. [8]. The central innovation of DML is the use of Neyman-orthogonal moments. This allows for flexible ML models to estimate nuisance functions without introducing first-order bias into the main parameter estimate. This robustness to regularisation bias is critical for reliable inference. The DML approach has been successfully applied to various causal inference problems. This includes estimating DTEs in randomised experiments [4]. We adapt and extend this powerful framework to the more complex case of an endogenous treatment requiring an IV, and critically examine the role of the ML model choice within this framework.

Fourth, this paper introduces and evaluates Kolmogorov-Arnold Networks (KANs) as nuisance learners for D-IV-LATE. KANs, introduced by Liu et al. [17], replace the fixed activation functions of traditional MLPs with learnable, spline-based activation functions on the network edges. This architectural innovation is inspired by the Kolmogorov-Arnold representation theorem and gives KANs a structured way to represent complex non-linearities. The theoretical properties of KANs, such as their approximation power in Besov spaces, have been explored by Kratsios & Furuya [15]. Those results motivate KANs as candidate nuisance learners, but they do not imply that the trained finite-sample estimator used here must dominate classical learners. Other related work includes probabilistic extensions like GP-KANs [5] and applications to ITE estimation such as KANITE [18]. Our contribution is to test this architecture in a DML D-IV-LATE pipeline and to document where its computational gains do and do not translate into statistical gains.

Finally, our work relates to the broader literature on semiparametric theory and efficient estimation [3, 19]. Through constructing estimators based on orthogonal influence functions within the DML framework, we aim for estimators that are robust to first-order errors in nuisance function estimation. This provides a crucial theoretical justification for our methods, bridging flexible ML estimation with rigorous econometric theory. It also highlights the importance of the properties of the chosen ML models (like KANs or Random Forests) in satisfying the underlying assumptions for valid inference.

3. Theoretical Framework and Identification

We adopt the potential outcomes framework to define our causal parameter of interest. Let Y_i be the observed outcome for unit i . Let $W_i \in \{0, 1\}$ be the indicator for a binary treatment that is potentially endogenous. We observe a vector of pre-treatment covariates $X_i \in \mathcal{X}$. To address the endogeneity of W_i , we assume the existence of a binary instrumental variable (IV), $Z_i \in \{0, 1\}$.

Following the convention in the IV literature, we define two sets of potential outcomes. First, let $W_i(z)$ be the potential treatment status for unit i if their instrument takes the value z . The observed treatment is then $W_i = W_i(Z_i)$. Second, let $Y_i(w, z)$ be the potential outcome for unit i if their treatment status were w and their instrument were z . The standard exclusion restriction assumes that the instrument Z_i only

affects the outcome Y_i through its effect on the treatment W_i . This implies that $Y_i(w, 1) = Y_i(w, 0)$ for $w \in \{0, 1\}$. We can therefore simplify the notation to $Y_i(w)$, which denotes the potential outcome for unit i under treatment w . The observed outcome is thus $Y_i = Y_i(W_i)$.

The data consist of n i.i.d. observations of (Y_i, W_i, Z_i, X_i) . Our goal is to estimate the causal effect of W on the distribution of Y .

3.1. Assumptions and Parameter of Interest

To identify the causal effect of W on the distribution of Y using the instrument Z , we rely on the following standard assumptions conditional on the covariates X .

Assumption 3.1 (Independence and Exclusion). The instrument Z_i is independent of the potential outcomes and potential treatment assignments, conditional on the covariates X_i

$$Z_i \perp\!\!\!\perp (Y_i(1), Y_i(0), W_i(1), W_i(0)) | X_i.$$

The exclusion restriction is embedded in our definition of $Y_i(w)$, which does not depend on Z_i .

Assumption 3.2 (Relevance). The instrument has a causal effect on the treatment assignment, conditional on X_i

$$\mathbb{P}(W_i = 1 | Z_i = 1, X_i) \neq \mathbb{P}(W_i = 1 | Z_i = 0, X_i).$$

This ensures that there is a non-zero proportion of individuals whose treatment status is affected by the instrument.

Assumption 3.3 (Monotonicity). The instrument affects the treatment decision in the same direction for all individuals. That is, for all i

$$W_i(1) \geq W_i(0).$$

This assumption rules out the existence of "defiers"—individuals who would take the treatment if unencouraged ($Z_i = 0$) but not if encouraged ($Z_i = 1$).

Under these assumptions, the instrument allows us to identify the treatment effect for the subpopulation of "compliers"—individuals who comply with the instrument, i.e., those for whom $W_i(1) > W_i(0)$.

Our parameter of interest is the Distributional Instrumental Variable LATE (D-IV-LATE).¹ This is the difference in the potential outcome distributions for the population of compliers. For any outcome level $y \in \mathcal{Y}$, the D-IV-LATE is defined as

$$\Delta(y) := \mathbb{P}(Y(1) \leq y | W(1) > W(0)) - \mathbb{P}(Y(0) \leq y | W(1) > W(0)). \quad (1)$$

3.2. Identification

Following the seminal work of Imbens & Angrist [12], the D-IV-LATE can be identified from the observed data. The key insight is that the difference in the conditional distribution of the outcome given the instrument can be written as a weighted average of the effects for compliers, always-takers, and never-takers. Under the assumptions, the effects for always-takers and never-takers are zero. This leaves only the effect for compliers.

¹We use D-IV-LATE to stand for *Distributional Instrumental Variable Local Average Treatment Effect*. This distinguishes our parameter from the "D-LATE" acronym used by Hoagland [11] for *Dynamic Local Average Treatment Effect*, a different concept focused on the evolution of average effects over time.

Without covariates, or after conditioning has already been absorbed into the relevant reduced forms, the D-IV-LATE is identified by the following Wald-type ratio.

$$\Delta(y) = \frac{\mathbb{E}[\mathbf{1}\{Y_i \leq y\} | Z_i = 1] - \mathbb{E}[\mathbf{1}\{Y_i \leq y\} | Z_i = 0]}{\mathbb{E}[W_i | Z_i = 1] - \mathbb{E}[W_i | Z_i = 0]}. \quad (2)$$

The numerator is the intent-to-treat (ITT) effect on the distributional outcome $\mathbf{1}\{Y \leq y\}$. The denominator is the ITT effect on the treatment uptake, which is also the proportion of compliers in the population. Under conditional independence given X , the covariate-adjusted target is obtained by integrating the conditional reduced-form contrasts over the covariate distribution.

$$\Delta(y) = \frac{\mathbb{E}_X[\mathbb{E}[\mathbf{1}\{Y_i \leq y\} | Z_i = 1, X_i] - \mathbb{E}[\mathbf{1}\{Y_i \leq y\} | Z_i = 0, X_i]]}{\mathbb{E}_X[\mathbb{E}[W_i | Z_i = 1, X_i] - \mathbb{E}[W_i | Z_i = 0, X_i]]}. \quad (3)$$

While this classic result provides a path to identification, a simple "plug-in" estimator based on this formula, where the conditional expectations are replaced by machine learning predictions, typically suffers from regularisation bias. The next section develops estimators that overcome this challenge.

4. Estimation Strategy

While Equation 3 identifies the D-IV-LATE as a ratio of reduced-form contrasts, a high-quality estimator still requires more than direct plug-in substitution. With flexible nuisance learners, naive plug-in estimation generally suffers from regularisation bias. The double/debiased machine learning (DML) strategy of Chernozhukov et al. [8] addresses that problem by constructing scores whose first-order sensitivity to nuisance-estimation error is zero at the truth.

4.1. Reduced-Form Nuisance Functions

For a fixed threshold y , the corrected estimator in this paper is built from the reduced-form nuisance bundle

$$\begin{aligned} g_z(y, x) &:= \mathbb{E}[\mathbf{1}\{Y \leq y\} | Z = z, X = x], & z \in \{0, 1\}, \\ m_z(x) &:= \mathbb{E}[W | Z = z, X = x], & z \in \{0, 1\}, \\ \pi(x) &:= \mathbb{P}(Z = 1 | X = x). \end{aligned}$$

The corresponding level parameters are

$$\begin{aligned} \alpha_0(y) &= \mathbb{E}[g_1(y, X) - g_0(y, X)], \\ \beta_0 &= \mathbb{E}[m_1(X) - m_0(X)], \end{aligned}$$

so that $\Delta_0(y) = \alpha_0(y)/\beta_0$ whenever $\beta_0 \neq 0$. This nuisance bundle lines up directly with the identification formula in Section 3: g_z is the reduced-form distribution of the observed outcome under instrument state z , and m_z is the reduced-form first stage under the same state.

4.2. Machine Learning Models for Nuisance Functions

4.2.1. Standard Machine Learning Models (e.g., Random Forests)

A common approach is to estimate nuisance functions using well-established supervised machine learning methods such as Random Forests, gradient boosting, or neural networks. Random Forests, for example, are

ensembles of decision trees that are robust to overfitting and can capture non-linear relationships in the data. They are widely used in applied causal inference due to their good empirical performance and availability in standard software packages. In the context of the D-IV-LATE estimator, Random Forests are used for the reduced-form nuisance bundle $(\hat{g}_0, \hat{g}_1, \hat{m}_0, \hat{m}_1, \hat{\pi})$.

4.2.2. Kolmogorov-Arnold Networks (KANs) as Candidate Nuisance Learners

While the DML framework provides robustness to first-order estimation errors, the choice of ML model for nuisance functions is far from inconsequential. We evaluate Kolmogorov-Arnold Networks (KANs) because their architecture is plausibly well suited to the repeated reduced-form regressions required by D-IV-LATE, not because the architecture alone guarantees better causal estimates than Random Forests or standard Multi-Layer Perceptrons (MLPs).

First, KANs provide a flexible approximation architecture. Unlike MLPs, which learn global, fixed-basis activation functions, KANs learn activation functions on each edge of the network. These are typically parameterised as B-splines [17]. This gives the learner a local representation of complex, highly non-linear relationships. The Kolmogorov-Arnold representation theorem, which inspires the KAN architecture, suggests that multivariate continuous functions can be represented as compositions of univariate functions. KANs operationalise this idea and are therefore natural candidates for approximating the reduced-form nuisance surfaces. Whether this architectural promise improves finite-sample D-IV-LATE estimation is an empirical question addressed by the benchmark below.

Second, recent theoretical work on KANs provides useful approximation results, especially for rich smoothness classes such as Besov spaces [15]. Those results make KANs attractive candidates for complex nuisance learning, but they do not by themselves establish the statistical convergence rates required for DML inference under the training pipeline used in this paper. We therefore use the KAN theory primarily as architectural motivation rather than as a completed nuisance-rate justification.

Third, KANs may be computationally efficient for repeated nuisance estimation. By learning edge-level activation functions, KANs can sometimes achieve useful approximation quality with relatively compact networks. This is attractive in the D-IV-LATE setting where we must estimate the reduced-form nuisance surfaces $g_0(y, x)$ and $g_1(y, x)$ over an outcome grid. The simulation evidence below shows that this computational motivation is borne out in runtime, even though it does not translate into broad RMSE dominance.

4.2.3. Nuisance Function Estimation with KANs: Implementation Details

The practical implementation of our KAN-D-IV-LATE estimator relies on the `efficient-kan` library² which is a PyTorch-based version designed for improved computational performance.

For estimating $\hat{\pi}(x)$, $\hat{m}_0(x)$, $\hat{m}_1(x)$, $\hat{g}_0(y, x)$, and $\hat{g}_1(y, x)$, we use KAN architectures specified by the `layers_hidden` parameter in `efficient-kan`. The locked KAN policy used for the corrected numerical evidence is `kan_width64_v1`, with a structure of `[input_dim, 64, 1]`. This structure corresponds to an input layer, one hidden layer with 64 neurons, and a single output neuron. The `input_dim` varies by nuisance function: it is d_X for $\hat{\pi}(x)$ and $d_X + 1$ for reduced-form nuisances that include the instrument indicator as an input. In the active implementation, $\hat{m}_0$ and $\hat{m}_1$ are obtained by fitting a single learner for $\mathbb{E}[W \mid Z, X]$ and evaluating it at $Z = 0$ and $Z = 1$; $\hat{g}_0$ and $\hat{g}_1$ are obtained analogously from a learner for $\mathbb{E}[\mathbf{1}\{Y \leq y\} \mid Z, X]$.

The locked KAN specification employs cubic splines with `grid_size` 4, AdamW with learning rate 1×10^{-3} , weight decay 1×10^{-4} , and an L1-type spline regularization penalty of 1×10^{-4} . It is trained for

²Our specific implementation is available at <https://github.com/shawcharles/efficient-kan>. The original KAN concept was introduced by Liu et al. [17].

25 optimization steps. The corrected empirical and inference reruns use the same locked KAN policy so that the numerical evidence is aligned across sections.

The `efficient-kan` library provides internal L1-type regularization on spline weights through its regularization-loss method, which we add directly to the training loss. For numerical stability, if the minority-class count for a binary nuisance target within a training fold falls below a prespecified threshold, KAN fitting is bypassed and the predictor is set to the majority-class value.

Cross-fitting varies by experiment. The main simulation benchmark uses $K = 3$ folds for computational tractability across the 27-scenario matrix, whereas the empirical application uses $K = 5$ folds. Section 7.2 reports the targeted KAN ablation over width, grid size, training steps, and regularization strength. That ablation selects the width-64 variant as the best KAN point-estimation configuration across its representative scenarios.

4.3. Orthogonal Level Scores for D-IV-LATE

The corrected estimator uses orthogonal level scores for the reduced-form numerator and denominator. Let $\eta(y) := (g_0(y, \cdot), g_1(y, \cdot), m_0, m_1, \pi)$. For the denominator, define

$$\phi_{\beta,i}(\eta) = m_1(X_i) - m_0(X_i) + \frac{Z_i}{\pi(X_i)}(W_i - m_1(X_i)) - \frac{1 - Z_i}{1 - \pi(X_i)}(W_i - m_0(X_i)). \quad (4)$$

For the numerator at threshold y , define

$$\begin{aligned} \phi_{\alpha,i}(y; \eta) &= g_1(y, X_i) - g_0(y, X_i) \\ &+ \frac{Z_i}{\pi(X_i)}(\mathbf{1}\{Y_i \leq y\} - g_1(y, X_i)) - \frac{1 - Z_i}{1 - \pi(X_i)}(\mathbf{1}\{Y_i \leq y\} - g_0(y, X_i)). \end{aligned} \quad (5)$$

Standard iterated-expectation arguments imply

$$\mathbb{E}[\phi_{\beta,i}(\eta_0)] = \beta_0, \quad \mathbb{E}[\phi_{\alpha,i}(y; \eta_0)] = \alpha_0(y),$$

so the D-IV-LATE remains the ratio $\Delta_0(y) = \alpha_0(y)/\beta_0$. The corresponding orthogonal ratio score is

$$\psi_{\Delta,i}(y; \Delta, \eta) = \phi_{\alpha,i}(y; \eta) - \Delta \phi_{\beta,i}(\eta),$$

whose moment vanishes at $(\Delta_0(y), \eta_0)$. In practice we estimate $\Delta_0(y)$ by the ratio of cross-fitted sample means,

$$\hat{\Delta}(y) = \frac{\frac{1}{n} \sum_{i=1}^n \hat{\phi}_{\alpha,i}(y)}{\frac{1}{n} \sum_{i=1}^n \hat{\phi}_{\beta,i}}. \quad (6)$$

This is the corrected estimator now implemented in the active code path.

4.4. Algorithm: Cross-Fitting

To implement the estimator in practice and prevent overfitting bias from the use of machine learning, we use a K -fold cross-fitting procedure. The steps are as follows. First, the n observations are randomly split into K disjoint folds, denoted I_k for $k = 1, \dots, K$. Next, for each fold $k \in \{1, \dots, K\}$, the data in all folds except k (the training set I_k^c) are used to estimate $\hat{\pi}_k(x)$, $\hat{m}_{0,k}(x)$, $\hat{m}_{1,k}(x)$, $\hat{g}_{0,k}(y, x)$, and $\hat{g}_{1,k}(y, x)$, using either Random Forests or KANs. In the active implementation, $\hat{m}_{0,k}$ and $\hat{m}_{1,k}$ come from a learner for

$\mathbb{E}[W \mid Z, X]$ evaluated at $Z = 0$ and $Z = 1$, and $\hat{g}_{0,k}$ and $\hat{g}_{1,k}$ come from a learner for $\mathbb{E}[\mathbf{1}\{Y \leq y\} \mid Z, X]$ evaluated at the same counterfactual instrument states. Subsequently, for each observation $i \in I_k$, the fold- k predictions are plugged into the score components

$$\begin{aligned}\hat{\phi}_{\beta,i} &= \hat{m}_{1,k}(X_i) - \hat{m}_{0,k}(X_i) \\ &\quad + \frac{Z_i}{\hat{\pi}_k(X_i)}(W_i - \hat{m}_{1,k}(X_i)) - \frac{1 - Z_i}{1 - \hat{\pi}_k(X_i)}(W_i - \hat{m}_{0,k}(X_i)), \\ \hat{\phi}_{\alpha,i}(y) &= \hat{g}_{1,k}(y, X_i) - \hat{g}_{0,k}(y, X_i) \\ &\quad + \frac{Z_i}{\hat{\pi}_k(X_i)}(\mathbf{1}\{Y_i \leq y\} - \hat{g}_{1,k}(y, X_i)) \\ &\quad - \frac{1 - Z_i}{1 - \hat{\pi}_k(X_i)}(\mathbf{1}\{Y_i \leq y\} - \hat{g}_{0,k}(y, X_i)).\end{aligned}$$

Finally, the estimate for each level y is computed as the ratio in Equation 6. This cross-fitting procedure ensures that for each observation, the nuisance function predictions are made by a model that was not trained on that observation. This avoids bias from overfitting and satisfies conditions required by asymptotic theory.

5. Asymptotic Theory for the Corrected Estimator

Assumption 5.1 (Overlap and First-Stage Strength). There exists $\epsilon > 0$ such that $\epsilon \leq \pi_0(X) \leq 1 - \epsilon$ almost surely, and the denominator satisfies $|\beta_0| \geq \epsilon$.

Assumption 5.2 (Cross-Fitted Nuisance Accuracy). For each fixed y , the nuisance estimators are cross-fitted and satisfy

$$\|\hat{\pi} - \pi_0\|_2 \sum_{z=0}^1 \|\hat{m}_z - m_{0,z}\|_2 = o_p(n^{-1/2}), \quad \|\hat{\pi} - \pi_0\|_2 \sum_{z=0}^1 \|\hat{g}_z(y, \cdot) - g_{0,z}(y, \cdot)\|_2 = o_p(n^{-1/2}),$$

with all nuisance estimates uniformly bounded in probability.

Assumption 5.2 is the standard product-rate requirement behind DML arguments: first-order nuisance error drops out by orthogonality, so only second-order products remain. The corrected level scores in Equations 4 and 5 are orthogonal with respect to perturbations of $(g_0, g_1, m_0, m_1, \pi)$ around the truth.

Theorem 5.1 (Pointwise Asymptotic Linearity). Under Assumptions 3.1–3.3, 5.1, and 5.2, for each fixed threshold y ,

$$\sqrt{n}(\hat{\Delta}(y) - \Delta_0(y)) = \frac{1}{\beta_0} \frac{1}{\sqrt{n}} \sum_{i=1}^n (\phi_{\alpha,i}(y; \eta_0) - \Delta_0(y) \phi_{\beta,i}(\eta_0)) + o_p(1).$$

Consequently,

$$\sqrt{n}(\hat{\Delta}(y) - \Delta_0(y)) \rightsquigarrow \mathcal{N}(0, \sigma^2(y)),$$

where

$$\sigma^2(y) = \frac{\text{Var}(\phi_{\alpha,i}(y; \eta_0) - \Delta_0(y) \phi_{\beta,i}(\eta_0))}{\beta_0^2}.$$

The variance formula used in the code is the sample analogue of the influence representation in Theorem 5.1. For fixed y , this yields standard pointwise Wald intervals once the corrected estimator is used throughout the pipeline. The theorem is intentionally pointwise rather than uniform over y ; simultaneous inference for an entire estimated curve requires stronger empirical-process conditions and is left to future work.

One practical consequence is immediate. The corrected estimator in Equations 4–6 is not numerically equivalent to the residualized score used in the earlier draft. The main simulation, ablation, empirical-core, and inference tables reported below have therefore been regenerated under the corrected estimator. Results from the earlier residualized-score code path, including older robustness artifacts, are not used as final evidence in this draft.

The paper-facing numerical artifacts are anchored to the corrected evidence root `kan-d-iv-late/results/corrected_2026_06_06`. Within that root, the simulation benchmark is backed by the full width-64 profile manifest, the empirical results by the core empirical manifest, and the inference results by the full inference manifest. These manifests record the command, package versions, repository commits, and output files used for the tables below.

6. Statistical Inference and Uncertainty Quantification

Theorem 5.1 supplies a pointwise asymptotic justification for Wald intervals under the corrected estimator. That does not eliminate the need for finite-sample validation. Flexible nuisance learners, denominator fragility, and repeated estimation over a y -grid can all degrade coverage in practice. We therefore study inference empirically as well as theoretically.

For each estimated D-IV-LATE curve we construct pointwise asymptotic intervals using the influence-function variance estimator implied by Theorem 5.1. We also construct a resampling-based alternative using a bootstrap that preserves the sample-splitting structure. The inferential comparison is conducted in three representative simulation scenarios: `smooth_low--weak--n500`, `baseline--medium--n1000`, and `complex_local--weak--n500`. Each model is evaluated over eight Monte Carlo replications, and coverage is averaged over the evaluation grid.

Table 1 reports the resulting coverage and interval-width summaries at the nominal 95% level. The validation profile is deliberately smaller than the point-estimation benchmark, so the coverage numbers should be read as diagnostic evidence rather than as a high-precision coverage study. A rough binomial calculation using the 64 pointwise coverage cells in each model-scenario pair gives standard errors between about 0.02 and 0.06, and this understates dependence across grid points within a replication. Even with that caveat, the pattern is clear. KAN-based intervals are too narrow and undercover badly. Depending on the scenario, asymptotic KAN coverage ranges from 0.500 to 0.813, and bootstrap KAN coverage ranges from 0.359 to 0.500. Random Forest intervals are materially better calibrated: asymptotic RF coverage ranges from 0.922 to 0.969. Bootstrap intervals do not repair the KAN undercoverage in this implementation; they generally make it worse.

These results substantially narrow the set of defensible claims. The current evidence does not support KANs as a validated inferential default. For the purposes of this paper, the empirical comparison is therefore interpreted as model-choice sensitivity in the point estimates, not as evidence that the KAN empirical curve supports reliable confidence statements. Where interval-based interpretation is necessary, RF asymptotic inference is the safer baseline among the procedures we tested. Uniform inference over y and more reliable interval construction for flexible nuisance learners remain open problems.

Table 1: Inference-validation summary across three representative scenarios. Coverage is averaged across the evaluation grid; the nominal target is 0.95. Each model-scenario pair uses eight Monte Carlo replications and 64 pointwise coverage cells.

Scenario	Model	Asym. cov.	Boot. cov.	Asym. width	Boot. width	Preferred
<code>smooth_low, weak, n=500</code>	KAN	0.813	0.500	0.927	0.841	Asymptotic
<code>smooth_low, weak, n=500</code>	RF	0.969	0.984	2.359	1.728	Asymptotic
<code>baseline, medium, n=1000</code>	KAN	0.500	0.359	0.182	0.134	Asymptotic
<code>baseline, medium, n=1000</code>	RF	0.969	0.922	0.162	0.180	Asymptotic
<code>complex_local, weak, n=500</code>	KAN	0.750	0.391	0.816	0.706	Asymptotic
<code>complex_local, weak, n=500</code>	RF	0.922	0.719	0.741	0.638	Asymptotic

7. Simulation Study

The simulation evidence in this paper serves two purposes. First, it evaluates point-estimation performance in a corrected estimand-aligned benchmark. Second, it tests whether the KAN results are robust to a limited but explicit hyperparameter search.

7.1. Simulation Design 1: Full Benchmark Matrix

The original simulation code path in this project mixed several distinct ideas: a treatment-assignment mechanism, a distributional truth object, and multiple partially incompatible score implementations. We replaced that setup with an explicit monotone principal-strata design so that the benchmark truth is a genuine complier D-IV-LATE rather than an ATE-style contrast mislabeled as LATE. For each scenario we generate a large truth bundle from the same data-generating process, compute the complier distributional effect on a fixed y -grid, and then compare cross-fitted RF and KAN estimators against that truth.

The full benchmark crosses three design families (`smooth_low`, `baseline`, and `complex_local`), three instrument-strength regimes (`weak`, `medium`, and `strong`), and three sample sizes ($n \in \{500, 1000, 2000\}$). Each scenario uses 50 Monte Carlo replications, 10 evaluation points in y , $K = 3$ cross-fitting folds, and a fixed truth bundle of size 50,000. The baseline model comparison is therefore a 27-scenario matrix. The corrected full-matrix aggregate is the locked-policy `kan_width64_v1` run; partial resume aggregates from intermediate runs are not used for the tables below.

Table 2: Full corrected simulation benchmark summary under the locked `kan_width64_v1` policy. Reported deltas are KAN minus RF.

Metric	Value
Number of scenarios	27
Replications per scenario	50
KAN lower integrated RMSE	1/27
KAN lower mean absolute error	0/27
KAN faster mean runtime	27/27
Mean integrated RMSE delta	0.073
Median integrated RMSE delta	0.060
Mean absolute-error delta	0.077
Mean runtime delta (seconds)	-3.169

The corrected performance pattern is a speed/accuracy tradeoff rather than a KAN dominance result. KAN is faster in every one of the 27 benchmark scenarios, with an average runtime advantage of 3.169 seconds. RF, however, has lower integrated RMSE in 26 of 27 scenarios and lower mean absolute error in all 27. The only KAN integrated-RMSE win is `complex_local--strong--n2000`, where KAN has integrated

RMSE 0.073 versus 0.278 for RF. Even in that cell, RF has slightly lower mean absolute error, so the result should be interpreted as an RF tail-instability diagnostic rather than broad KAN accuracy superiority.

Table 3 breaks the result down by design family. The RF accuracy advantage is not confined to one design: KAN has zero integrated-RMSE wins in the **baseline** and **smooth_low** families and one win in **complex_local**. The runtime advantage is similarly broad across the design families.

Table 3: Corrected simulation benchmark by design family. Reported deltas are KAN minus RF.

Design	Cells	KAN RMSE wins	KAN MAE wins	Mean RMSE Δ	Runtime Δ
baseline	9	0	0	0.053	-2.986
complex_local	9	1	0	0.047	-3.522
smooth_low	9	0	0	0.120	-2.999

7.2. Simulation Design 2: Targeted KAN Ablation

To choose a KAN policy for the corrected reruns, we run a restricted ablation on three representative scenarios: **baseline--weak--n2000**, **complex_local--weak--n2000**, and **complex_local--strong--n2000**. The ablation varies network width, spline grid size, training steps, and regularization strength. Selection is based on average integrated RMSE, with ties broken by fragile denominator share and then runtime.

Table 4: Targeted KAN ablation across three representative scenarios. Lower is better for RMSE and runtime.

KAN configuration	Mean int. RMSE	Mean abs. error	Mean runtime
kan_width64_v1	0.194	0.170	4.986
kan_steps50_v1	0.198	0.173	9.245
kan_grid3_v1	0.251	0.223	4.905
kan_grid6_v1	0.254	0.226	4.955
kan_core_v1	0.255	0.226	5.104
kan_reg1e-5_v1	0.255	0.227	4.744

The ablation selects **kan_width64_v1**. The top two configurations are close in average integrated RMSE, but the 50-step variant is materially slower. This is an internal KAN policy lock, not a claim that KAN dominates RF. We therefore use the width-64, 25-step KAN as the locked KAN policy for the corrected simulation matrix, empirical core rerun, and inference validation.

8. Empirical Application: The Distributional Impact of 401(k) Participation on Financial Assets

We next study whether the simulation differences between RF and KAN nuisance learning matter in a standard empirical application. The setting is the distributional effect of 401(k) participation on net financial assets in the 1991 Survey of Income and Program Participation (SIPP).

8.1. Context and Data

We use the same SIPP sample analyzed by Chernozhukov & Hansen [6] and later work on 401(k) participation, consisting of 9,915 household reference persons. The outcome is net financial assets (**net_tfa**), the treatment is 401(k) participation (**p401**), and the instrument is 401(k) eligibility (**e401**). Covariates include income, age, education, marital status, family size, two-earner status, defined-benefit pension status, IRA participation, and homeownership. The empirical pipeline estimates the D-IV-LATE curve over 30 grid points unless otherwise noted.

8.2. Instrument and First Stage

In the corrected core raw specification, the sample first-stage effect is 0.705, the first-stage F -statistic is 12,594.96, and the corresponding partial R^2 is 0.560. These diagnostics make weak-instrument concerns secondary in this application. The more relevant issue is how the nuisance learner behaves when translating that strong first stage into a distributional effect curve.

8.3. Results and Comparative Discussion

The corrected core empirical comparison shows substantial divergence between RF and KAN, but that divergence must be interpreted together with the nuisance diagnostics. In the raw 30-point specification, the mean absolute gap between the two estimated curves is 0.624, the maximum pointwise gap is 2.989, and the sign-disagreement share is 0.767. KAN is also faster in this core empirical run, taking 71.39 seconds versus 102.23 seconds for RF. At the same time, the KAN propensity learner places 99.9% of observations outside the $[0.05, 0.95]$ interval, compared with 7.6% for RF. The empirical KAN curve is therefore best read as a sensitivity signal about nuisance geometry rather than as a stable standalone causal estimate.

8.4. Diagnostic Analysis

The corrected Phase 5 rerun refreshed the core empirical specification only. Earlier robustness runs used a pre-refresh numerical path and should not be quoted as final evidence. Table 5 therefore reports only the corrected core empirical divergence and records the main diagnostics needed for the next robustness-refresh step.

Table 5: Corrected empirical RF-KAN divergence and propensity diagnostics in the core specification.

Spec.	Mean gap	Max gap	Sign dis.	KAN outside	RF outside	KAN sec.	RF sec.
<code>core_raw_30</code>	0.624	2.989	0.767	0.999	0.076	71.39	102.23

Two findings stand out from the corrected core run. First, RF-KAN disagreement is large enough that model choice can change the sign and magnitude of point estimates along the estimated D-IV-LATE curve. Second, the empirical runtime ranking is consistent with the simulation matrix: the locked KAN policy is faster than RF in this implementation. Those findings do not by themselves validate the KAN empirical curve, because the propensity diagnostics indicate severe boundary concentration for the KAN nuisance learner.

The diagnostics also reveal why nuisance-model behavior matters. In the corrected raw core specification, 9,901 of 9,915 KAN instrument-propensity estimates fall outside $[0.05, 0.95]$, compared with 757 for RF. This is not a simple first-stage weakness story: the sample first stage is 0.705, with a first-stage F -statistic of 12,594.96 and partial R^2 of 0.560. Rather, it is a nuisance-geometry story in which different learners induce materially different orthogonal-score behavior even under the same identifying design. The grid, standardization, trimming, and probability-calibration robustness checks should be rerun under the corrected locked policy before they are used as final manuscript evidence.

The empirical application therefore reinforces the broader model-choice message, but with a sharper caveat. Flexible nuisance learning can change point-estimate conclusions in D-IV-LATE work. At the same time, the inference results in Section 6 and the overlap diagnostics in Table 5 imply that these KAN-based empirical curves should currently be interpreted as diagnostic sensitivity evidence rather than as validated inferential objects.

9. Caveat Emptor: The Role of Nuisance Function Estimators and Model Choice

The evidence in this paper changes what the relevant warning should be. The old caution was that model choice might matter. The stronger conclusion is that different components of performance matter differently. Point estimation, empirical shape recovery, numerical stability, and interval calibration do not necessarily rank models in the same order.

For point estimation, the corrected results are not uniformly favorable to KANs. In the corrected 27-scenario simulation matrix, the locked KAN policy is faster in every scenario, but RF has lower integrated RMSE in 26 scenarios and lower mean absolute error in all 27. The KAN ablation is still useful because it selects a disciplined KAN policy before the corrected reruns, but it does not overturn the RF accuracy advantage in the full matrix. For empirical interpretation, KAN and RF produce materially different D-IV-LATE curves in the 401(k) application. That makes nuisance-model choice a first-order substantive issue rather than an implementation detail.

For inference, the ranking changes. KAN intervals are not calibrated well enough to support primary claims. RF asymptotic intervals are not a universal solution, but they are materially better behaved in the validation study. The practical implication is that applied researchers should not treat a strong point-estimation result as evidence of inferential validity. In flexible DML pipelines, those are separate questions.

The appropriate “caveat emptor” is therefore not anti-machine-learning. It is methodological discipline. Researchers should benchmark nuisance learners under a truth-aligned design, document how robust the empirical curve is to preprocessing and tuning choices, and validate interval procedures directly before relying on them. Under the current evidence, KANs are best viewed as computationally attractive D-IV-LATE nuisance learners whose accuracy and inferential behavior remain problem-dependent.

10. Conclusion

This paper studies model choice for D-IV-LATE through a deliberately evidence-led sequence: a corrected simulation benchmark, a targeted KAN ablation, an empirical application, and an explicit inference validation exercise. Taken together, these results support a mixed conclusion.

For point estimation, KANs are computationally promising but not accuracy-dominant in the corrected benchmark. They are faster than RF across the full 27-scenario matrix, but RF has lower integrated RMSE in 26 scenarios and lower mean absolute error in all 27. The ablation selects a width-64 KAN as the best KAN variant among the configurations tested. In the 401(k) application, nuisance-model choice remains substantively important, with RF and KAN producing materially different D-IV-LATE curves in the corrected core specification. The KAN empirical curve, however, is accompanied by severe propensity-boundary concentration and should be treated as sensitivity evidence until calibration and robustness checks are refreshed.

For inference, however, the evidence is unfavorable to KANs. The validation study shows severe undercoverage for KAN-based pointwise intervals under both asymptotic and bootstrap constructions. RF asymptotic intervals are materially better calibrated in the validation design. The strongest defensible interpretation is therefore that KANs can be computationally useful nuisance learners, but the current implementation is not yet ready to serve as a default accuracy or inferential layer.

That mixed verdict is, in our view, the strongest publishable result. It is methodologically informative, empirically relevant, and harder to dismiss than a blanket claim of KAN superiority. The corrected estimator now rests on a genuine reduced-form orthogonal-score derivation, and the main numerical evidence has been

regenerated under that corrected code path. Future work should therefore focus not only on when KANs can beat classical learners on point RMSE, but also on how to pair flexible nuisance learning with reliable uncertainty quantification in semiparametric distributional-IV settings.

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